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(54) **REFRIGERATOR WITH MACHINE ROOM
OF WHICH COVER HAS ATTENUATED
VIBRATION**

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(57) **ABSTRACT**

A refrigerator includes a machine room provided at a lower portion of a main body; an opening provided on one side of the main body to allow access to the machine room; a cover disposed on one side of the main body to cover the opening; a plurality of cover catches provided on a lower portion of the cover; an emboss provided between the plurality of cover catches on the lower portion of the cover; and a plurality of main body hooks provided below the opening on one side of the main body such that locations of the plurality of main body hooks and the plurality of cover catches of the cover correspond to each other.

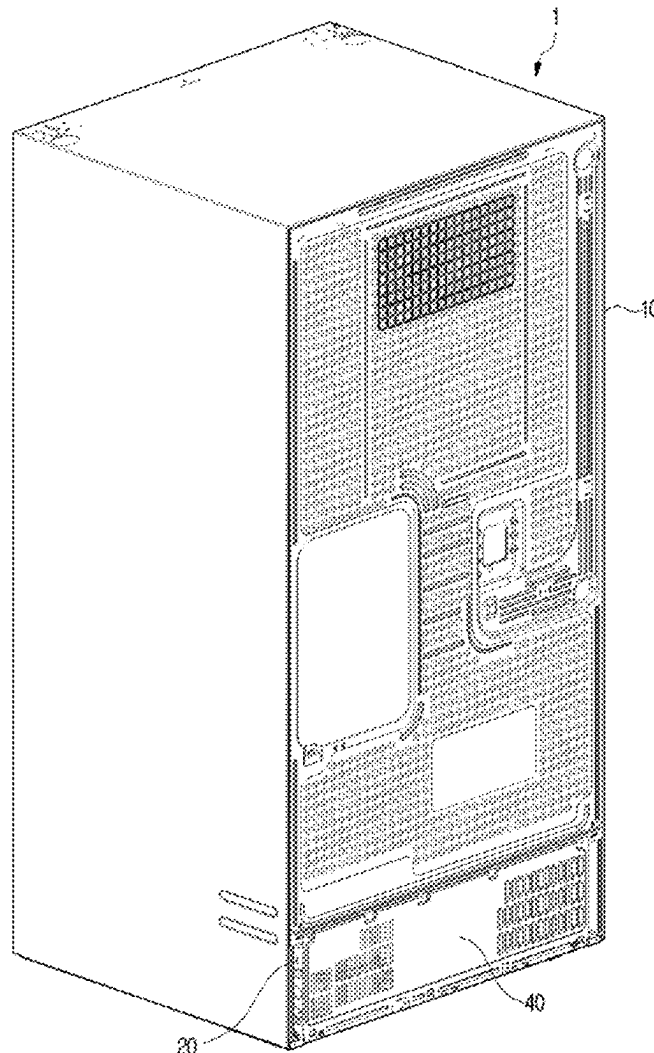


FIG. 1

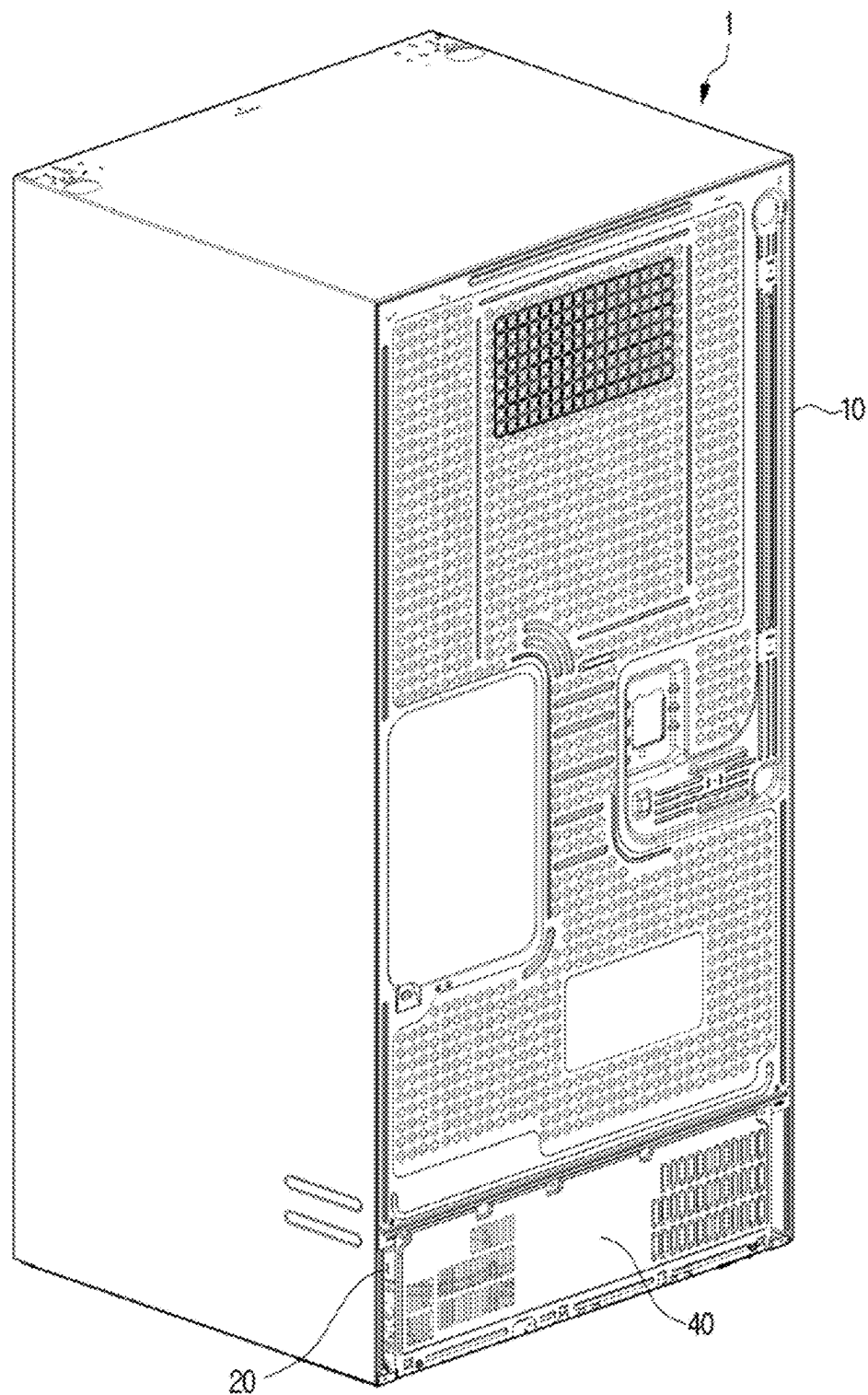


FIG. 2

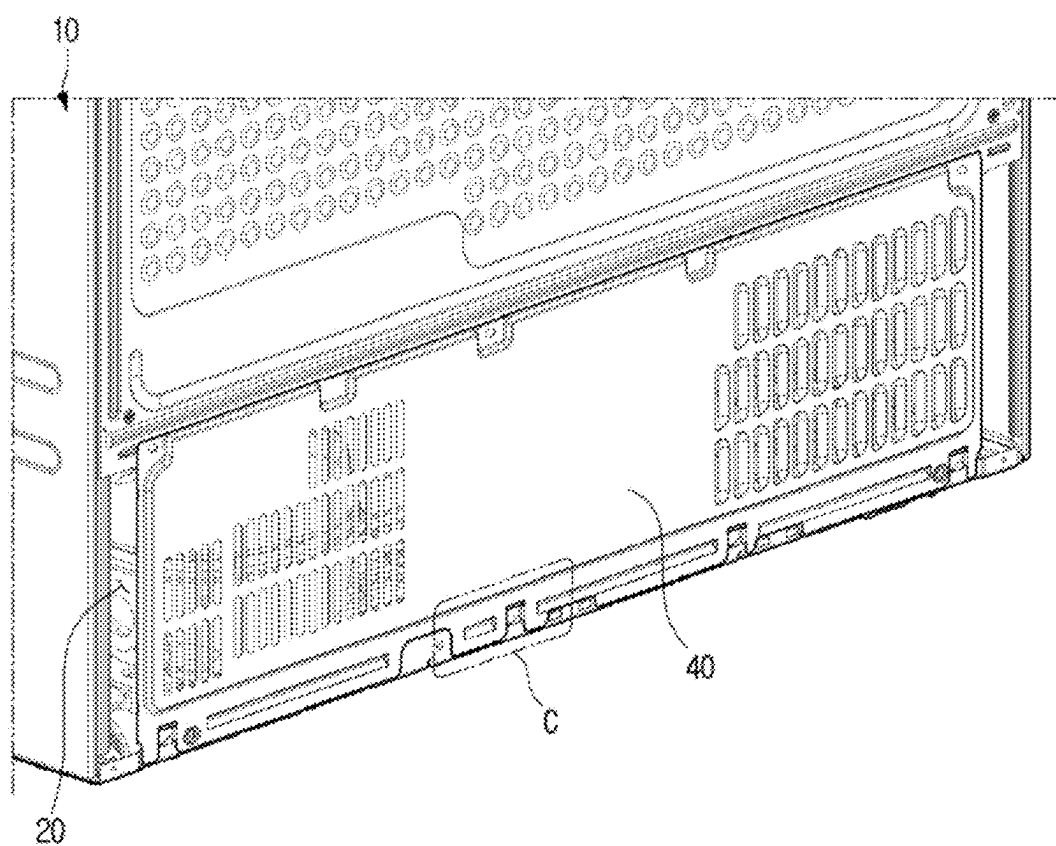


FIG. 3

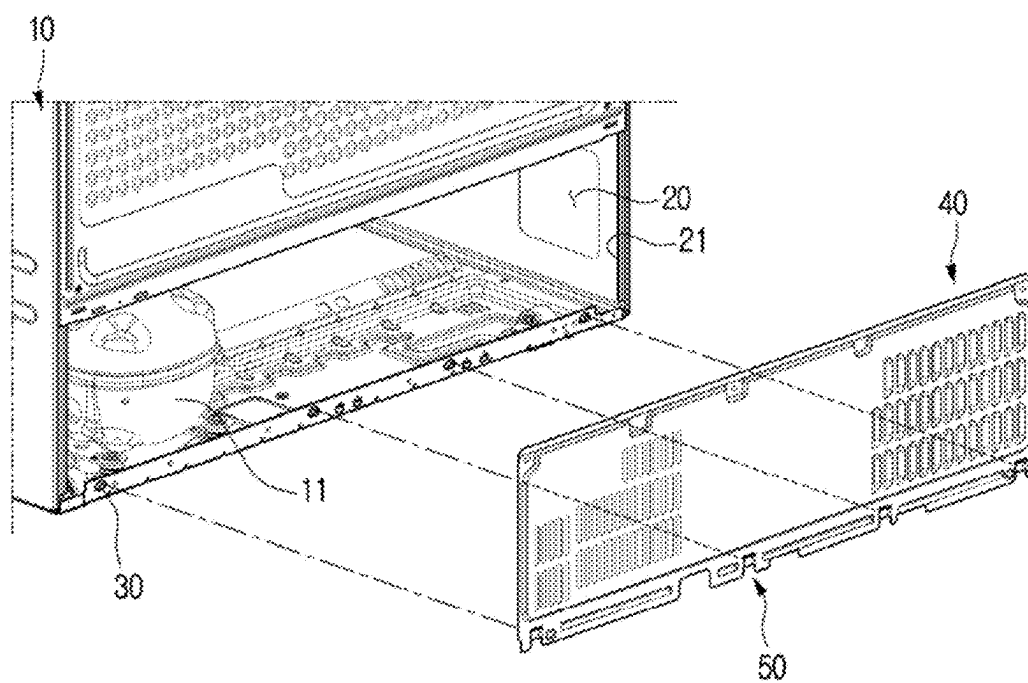


FIG. 4

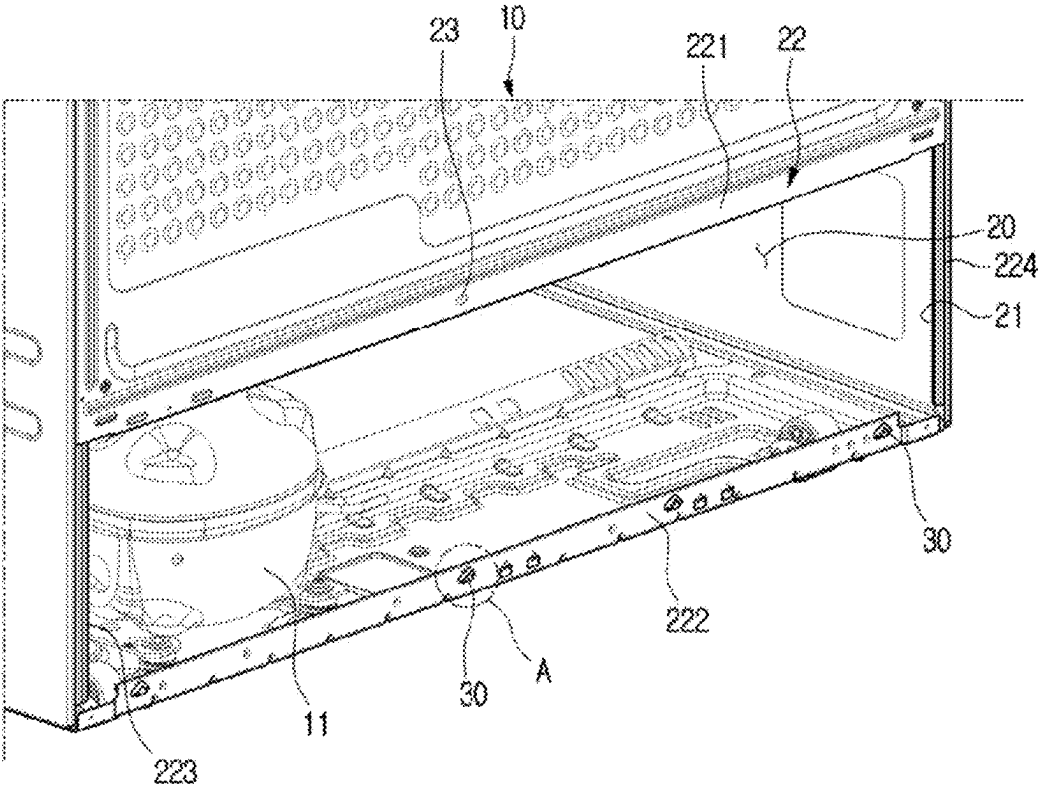
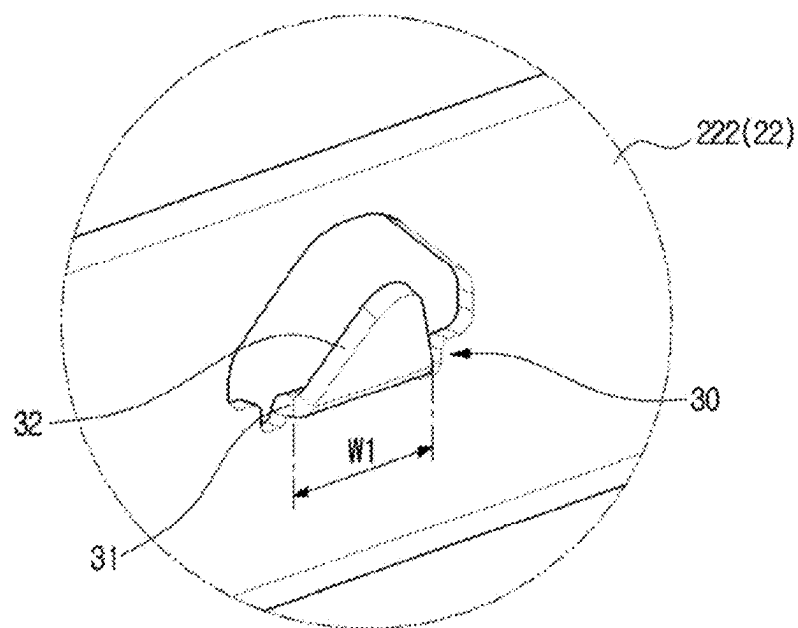
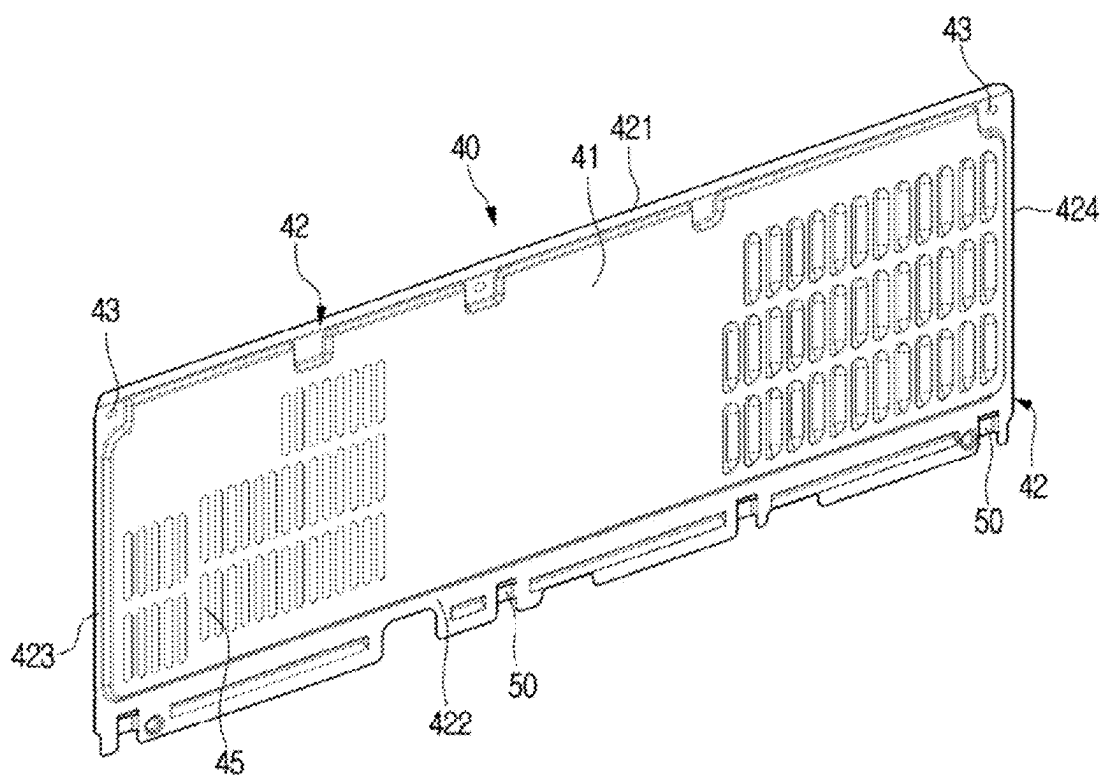


FIG. 5





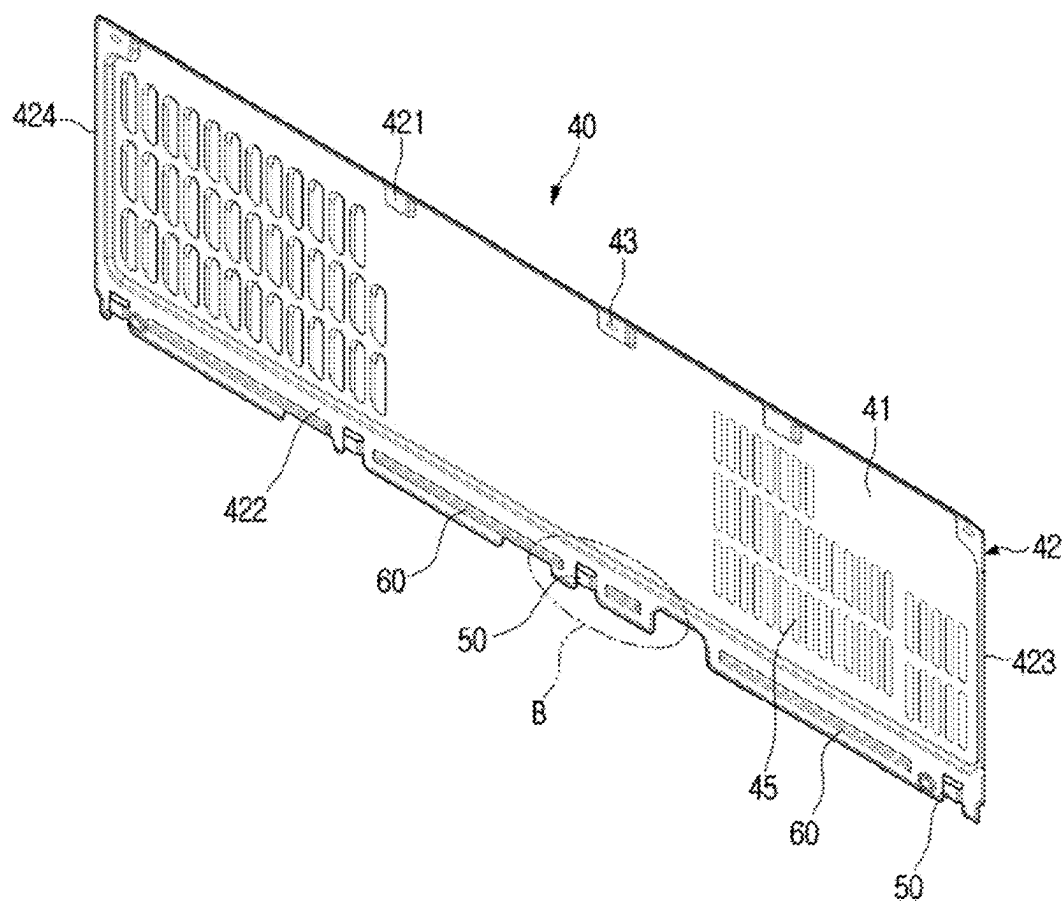
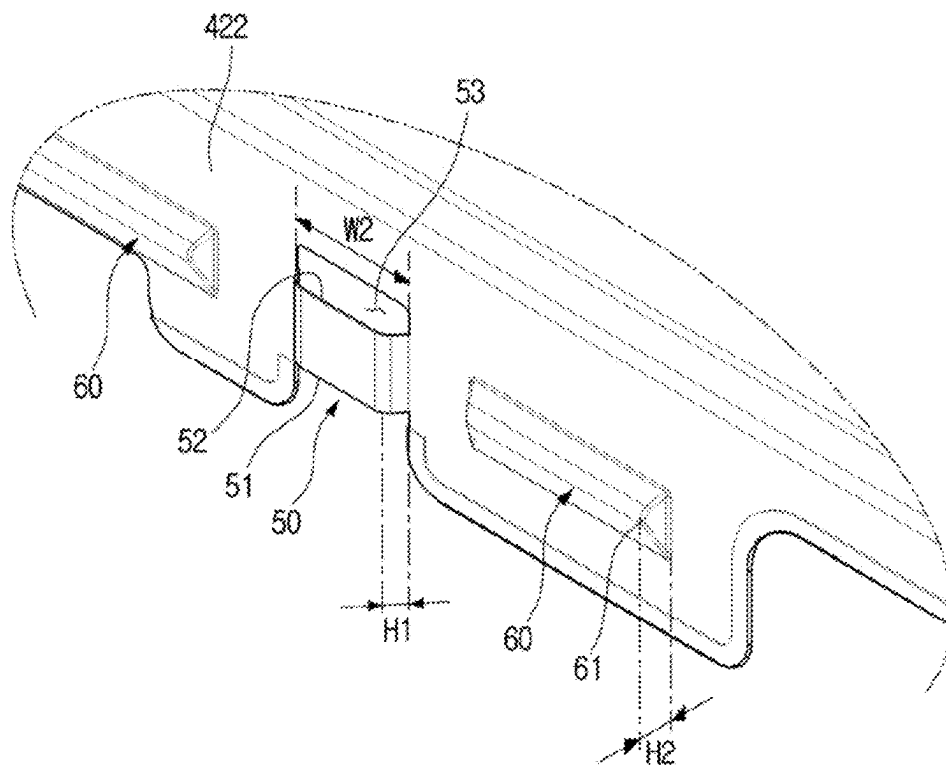


FIG. 8



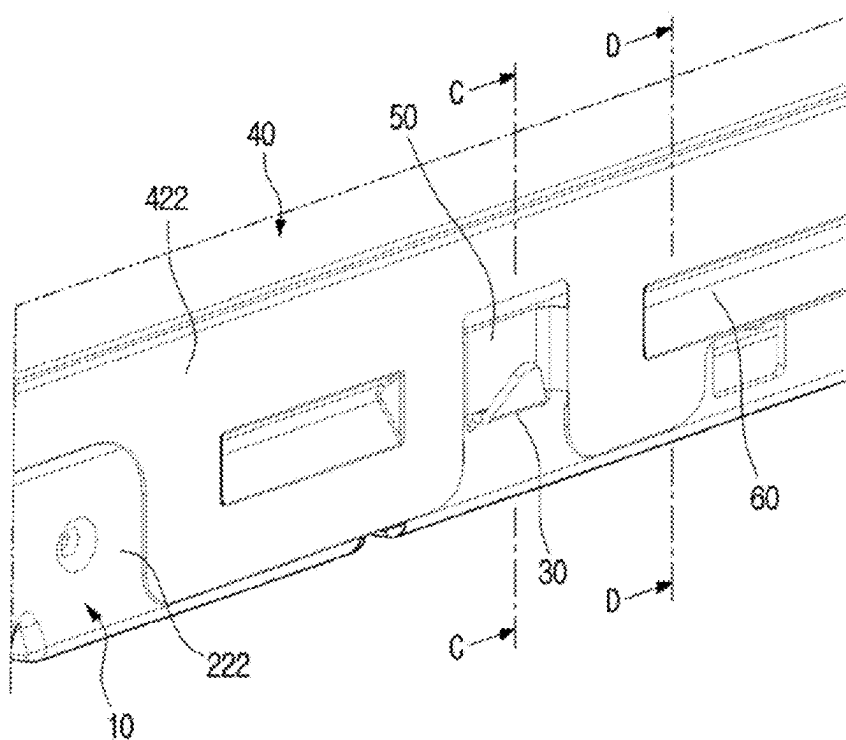


FIG. 10

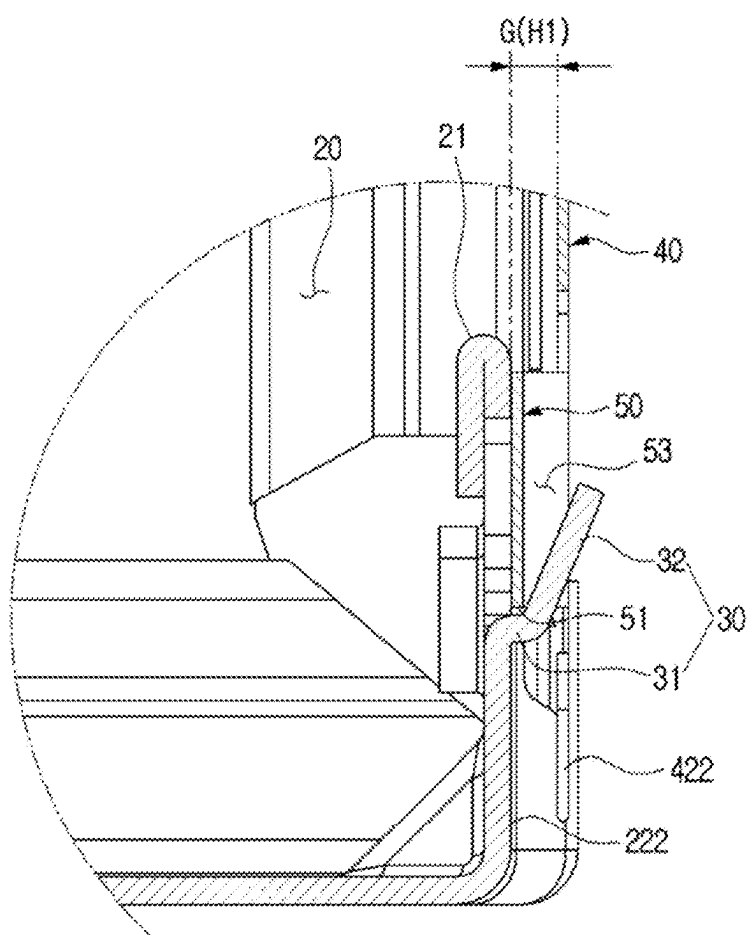


FIG. 11

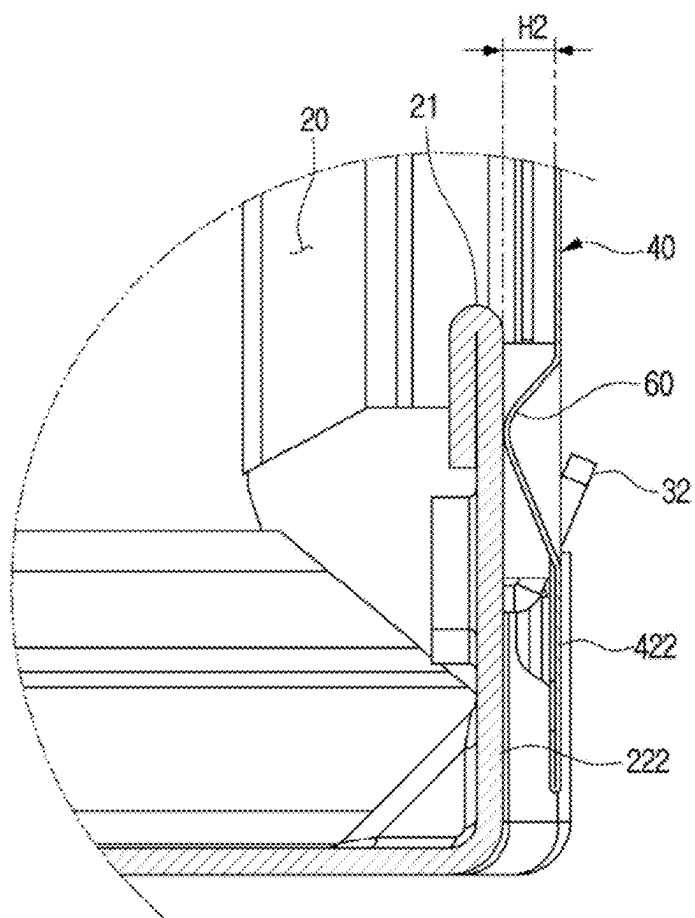


FIG. 12A

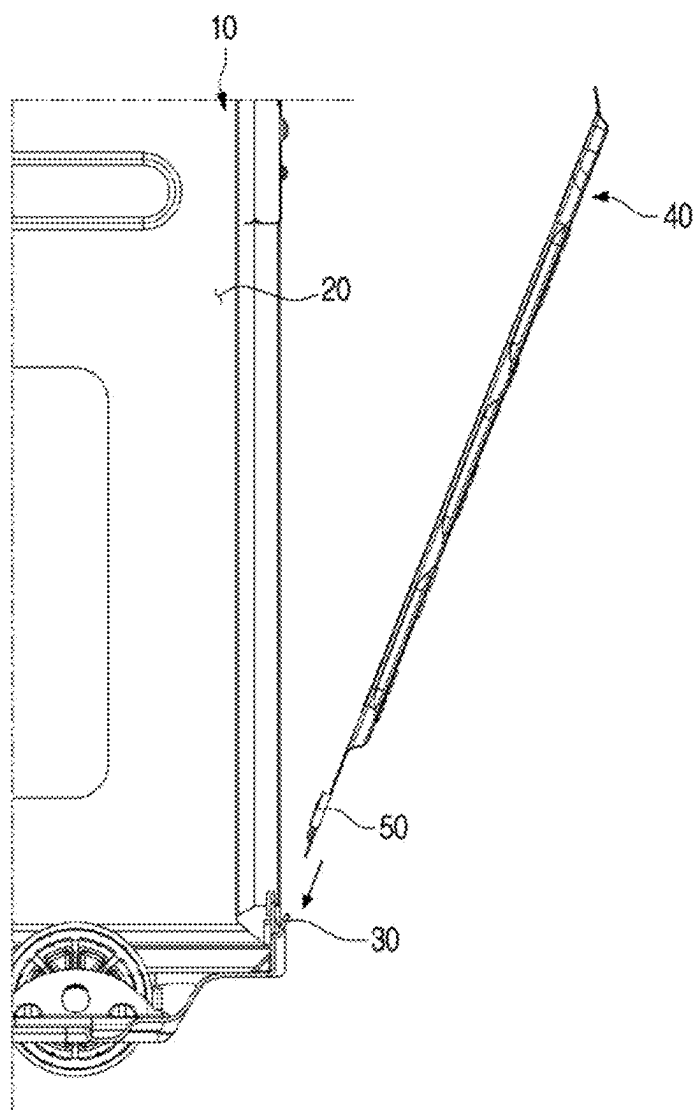


FIG. 12B

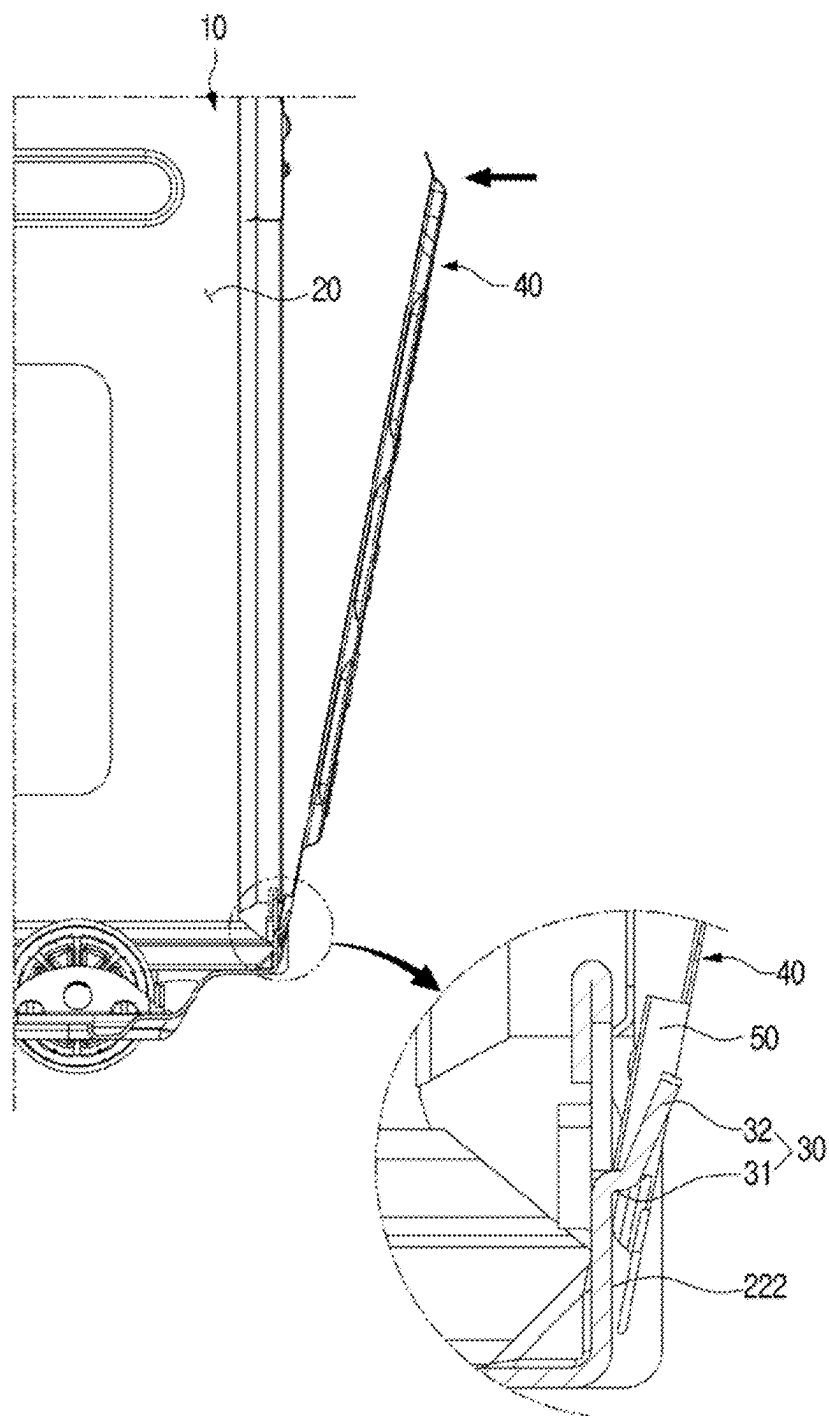


FIG. 12C

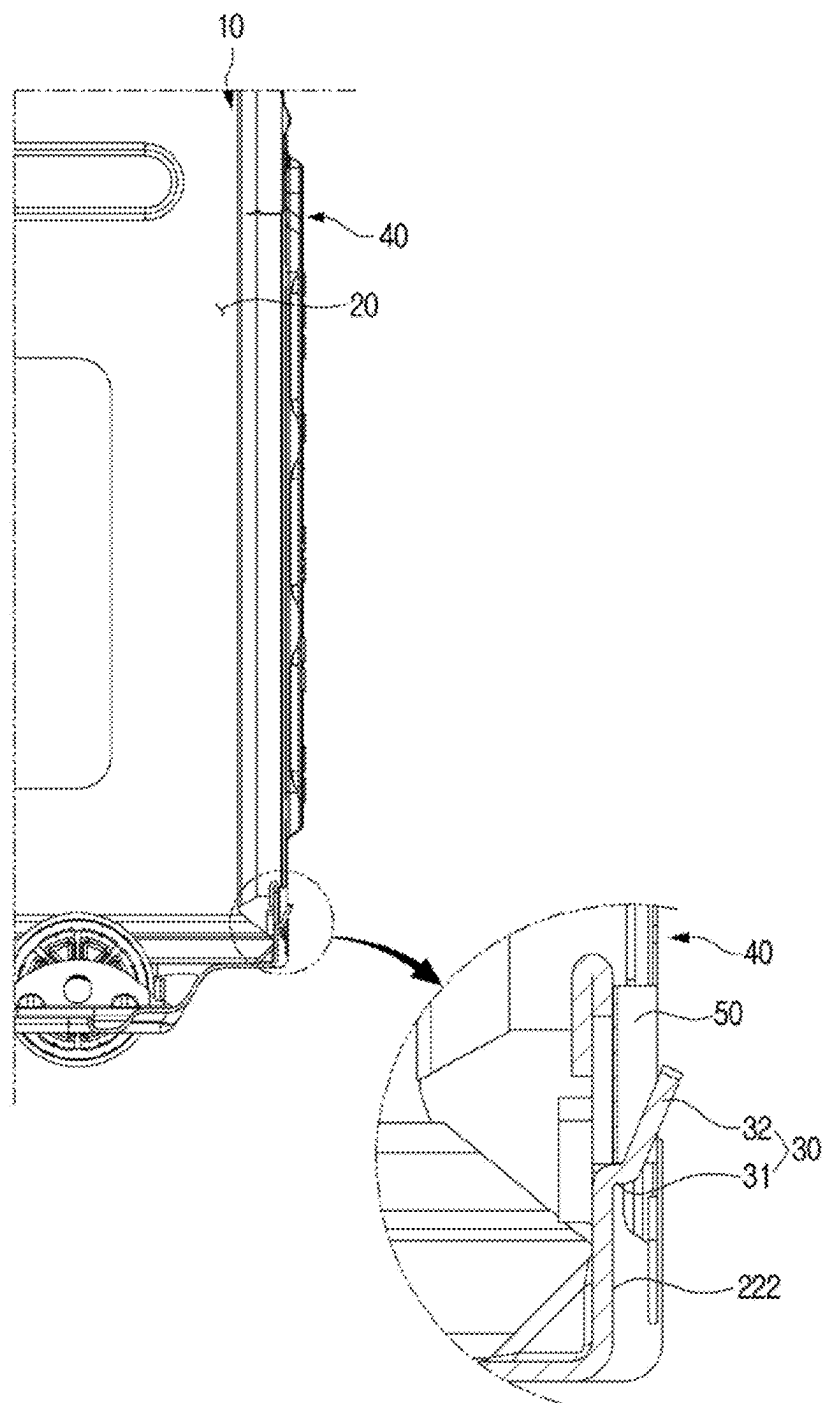


FIG. 13

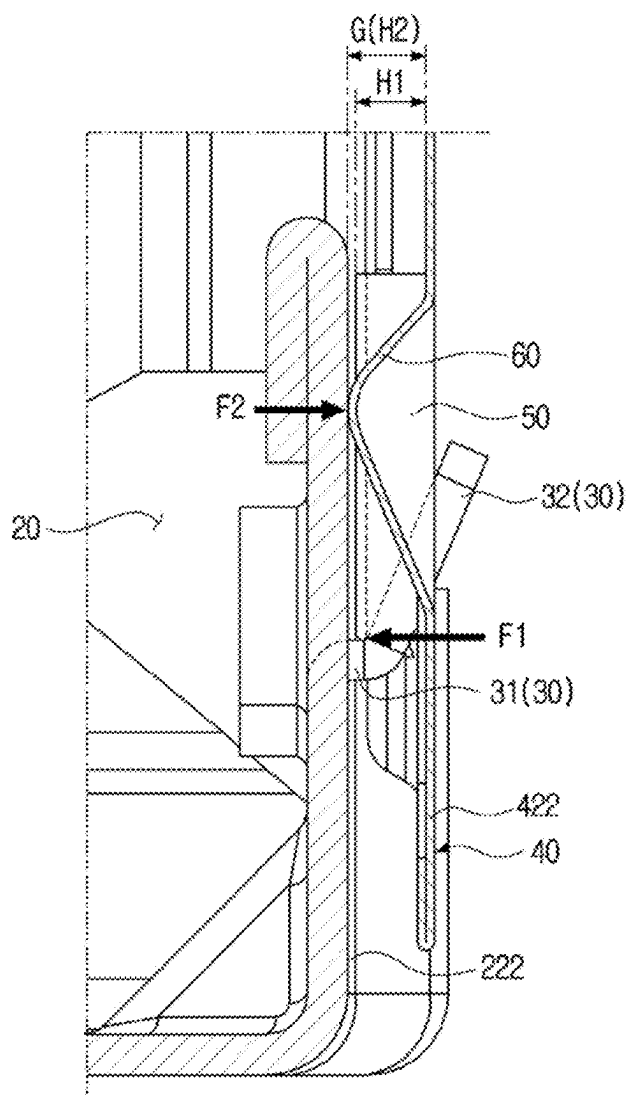


FIG. 14

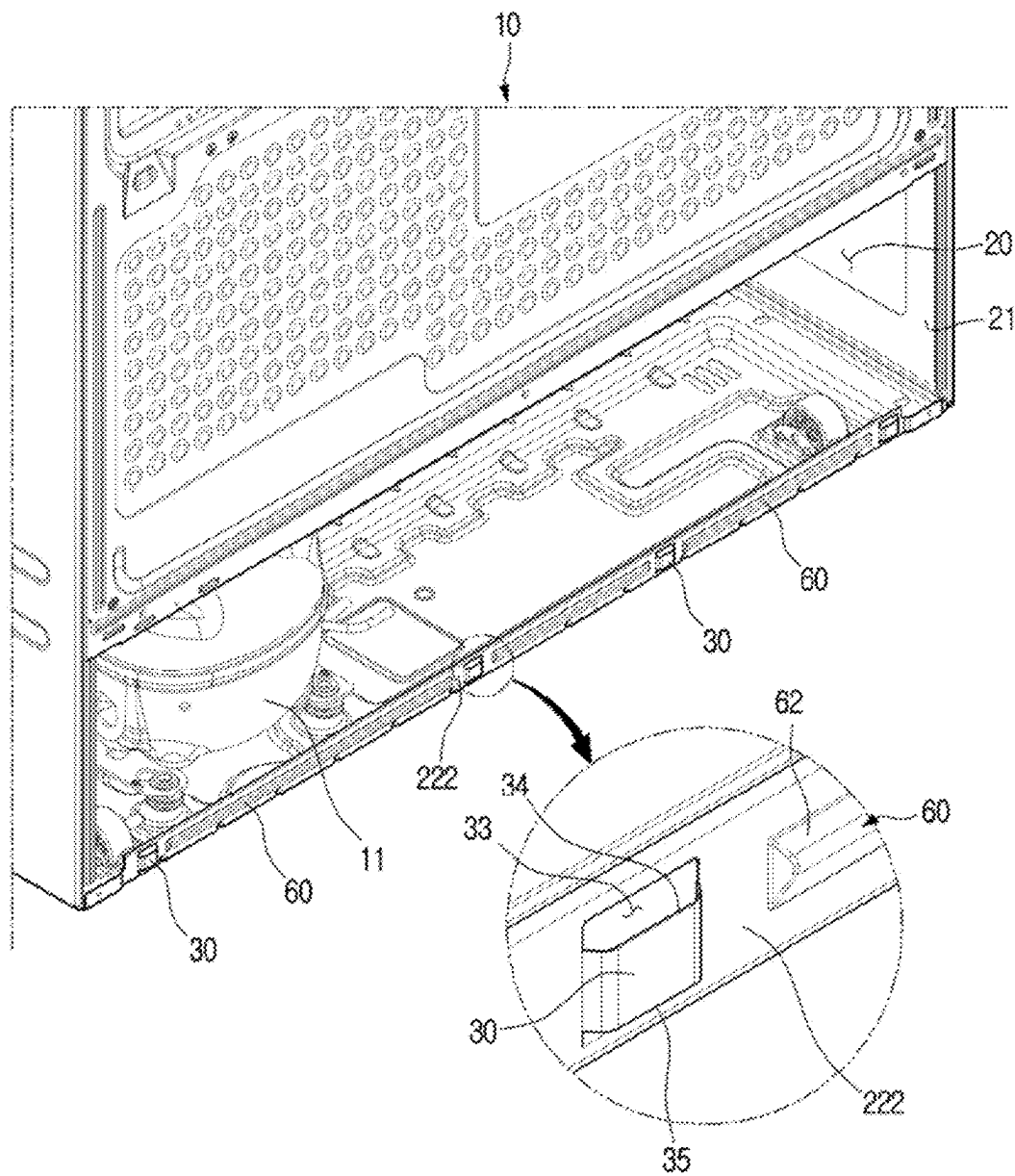
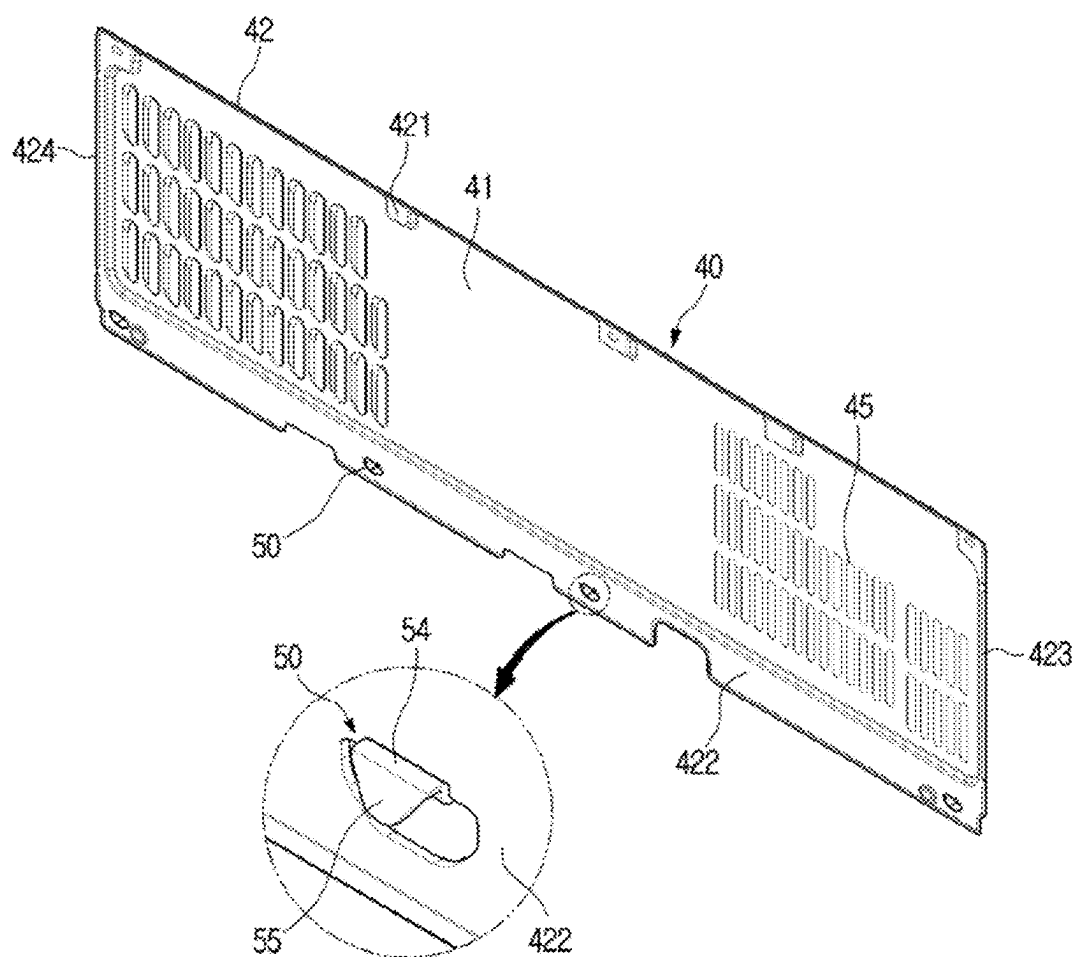


FIG. 15



REFRIGERATOR WITH MACHINE ROOM OF WHICH COVER HAS ATTENUATED VIBRATION

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation application is a continuation application, under 35 U.S.C. § 111(a), of international application No. PCT/KR2023/018583, filed Nov. 17, 2023, which claims priority under 35 U. S. C. § 119 to Korean Patent Application No. 10-2022-0174222, filed Dec. 13, 2023, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a refrigerator, and more particularly, to a refrigerator capable of damping vibration of a cover covering a machine room.

BACKGROUND ART

[0003] A refrigerator is a device that cools food or stores it at low temperature to prevent spoilage or deterioration of food.

[0004] The refrigerator may include a storage space that accommodates food and a cooling device that cools the storage space.

[0005] Generally, a refrigeration cycle device that generates cold air using a refrigeration cycle is used as the cooling device.

[0006] The refrigeration cycle device may generate cold air by circulating a refrigerant through a closed circuit including a compressor, a condenser, an expansion device, and an evaporator.

[0007] Therefore, the refrigerator may include a machine room in which the compressor and the condenser are accommodated. An opening communicating with the machine room may be provided on one side of the refrigerator. A cover may be detachably disposed in the opening of the machine room.

DISCLOSURE OF INVENTION

Technical Problem

[0008] The purpose of the disclosure is to provide a refrigerator capable of preventing or minimizing vibration noise of a cover caused by vibration of a compressor disposed in a machine room.

[0009] In addition, the purpose of the disclosure is to provide a refrigerator capable of reducing the vibration-proof cost for preventing vibration noise of a cover.

Technical Solution

[0010] According to an embodiment of the disclosure, a refrigerator may include: a machine room to be provided at a lower portion of a main body; an opening to be provided on one side of the main body to allow access to the machine room; a cover to be disposed on one side of the main body to cover the opening; a plurality of cover catches to be provided on a lower portion of the cover; an emboss to be provided between the plurality of cover catches on the lower portion of the cover; and a plurality of main body hooks to be provided on the one side of the main body below the

opening such that while the plurality of main body hooks are provided on the side of the main body, locations of the plurality of main body hooks and the plurality of cover catches of the cover.

[0011] While the opening of the machine room being covered with the cover, the plurality of cover catches may be caught on the plurality of main body hooks, and the emboss may contact the one side of the main body on which the plurality of main body hooks are formed.

[0012] A height of the emboss may be greater than a gap between the cover and the one side of the main body.

[0013] Each of the plurality of main body hooks may be formed to be inclined upwardly on one side of the main body.

[0014] The emboss may include a lower inclined surface formed to be inclined upwardly and facing a lower end of the cover.

[0015] The emboss may be formed to have an arc cross-section or a triangular cross-section.

[0016] The emboss may be among a plurality of embosses arranged in a straight line.

[0017] The cover may be formed as a rectangular flat plate. The plurality of cover catches may be provided on a lower portion of the cover. Each of the plurality of cover catches may be formed in a protrusion shape protruding from one side of the cover. A catch groove into which the main body hook is insertable may be provided at a lower end of the cover catch.

[0018] The emboss may be formed to protrude from one side of the cover in a same direction as the plurality of cover catches.

[0019] A height of the emboss may be higher than a height of the plurality of cover catches.

[0020] The refrigerator may further include: a fastening member formed to fasten an upper portion of the cover to one side of the main body above the opening.

[0021] The main body may include an upper support part, a lower support part, a left support part, and a right support part provided at an edge of the opening. The plurality of main body hooks may be provided on the lower support part. A plurality of screw holes may be provided on the upper support part.

[0022] The cover may include an upper fixing portion and a lower fixing portion. The plurality of cover catches may be provided on the lower fixing portion. A plurality of bolt holes corresponding to the plurality of screw holes of the upper support part of the main body 10 may be provided on the upper fixing portion.

[0023] According to an embodiment of the disclosure, a refrigerator may include: a machine room to be provided at a lower portion of a main body; an opening to be provided on one side of the main body to allow access to the machine room; a cover to be disposed on one side of the main body to cover the opening; a plurality of main body hooks to be provided on one side of the main body below the opening; an emboss to be provided on one side of the main body between the plurality of main body hooks and below the opening; and a plurality of cover catches to be provided on a lower portion of the cover such that while the cover is disposed on the one side of the main body, locations of the cover catches and the plurality of main body hooks of the main body correspond to each other.

[0024] Each of the plurality of cover catches may be formed to be inclined downward on a lower portion of the

cover. Each of the plurality of main body hooks may be formed in a protrusion shape protruding from one side of the main body. A catch groove into which the cover catch is insertable may be provided on an upper surface of the main body hook.

BRIEF DESCRIPTION OF DRAWINGS

[0025] These and/or other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0026] FIG. 1 is a rear perspective view illustrating a refrigerator 1 according to an embodiment of the disclosure.

[0027] FIG. 2 is a partial enlarged perspective view illustrating the refrigerator 1 of FIG. 1 according to an embodiment of the disclosure.

[0028] FIG. 3 is a perspective view illustrating a state in which a cover 40 is separated from the refrigerator 1 of FIG. 2 according to an embodiment of the disclosure.

[0029] FIG. 4 is a partial perspective view illustrating a machine room of a refrigerator according to an embodiment of the disclosure.

[0030] FIG. 5 is an enlarged perspective view of part A of FIG. 4 according to an embodiment of the disclosure.

[0031] FIG. 6 is a perspective view illustrating a cover configured to cover an opening of a machine room of a refrigerator according to an embodiment of the disclosure.

[0032] FIG. 7 is a perspective view illustrating the cover of FIG. 6 as viewed from the rear according to an embodiment of the disclosure.

[0033] FIG. 8 is an enlarged perspective view of part B of FIG. 7 according to an embodiment of the disclosure.

[0034] FIG. 9 is an enlarged perspective view of part C of FIG. 2 according to an embodiment of the disclosure.

[0035] FIG. 10 is a cross-sectional view taken along line C-C of FIG. 9 according to an embodiment of the disclosure.

[0036] FIG. 11 is a cross-sectional view taken along line D-D of FIG. 9 according to an embodiment of the disclosure.

[0037] FIG. 12A is a partial cross-sectional view illustrating a state before assembling a cover to a machine room of a refrigerator according to an embodiment of the disclosure.

[0038] FIG. 12B is a partial cross-sectional view illustrating a state in which a cover catch of a cover is coupled to a main body hook of a machine room of a refrigerator according to an embodiment of the disclosure.

[0039] FIG. 12C is a partial cross-sectional view illustrating a state in which a cover is assembled to a machine room of a refrigerator according to an embodiment of the disclosure.

[0040] FIG. 13 is a view illustrating forces acting between a machine room and a cover when the cover is disposed in the machine room of a refrigerator according to an embodiment of the disclosure.

[0041] FIG. 14 is a perspective view illustrating another example of a machine room of a refrigerator according to an embodiment of the disclosure.

[0042] FIG. 15 is a perspective view illustrating a cover disposed in the machine room of the refrigerator of FIG. 14 according to an embodiment of the disclosure.

BEST MODE FOR CARRYING OUT THE INVENTION

[0043] Since the embodiments of the disclosure can apply various transformations and have various embodiments, specific embodiments will be illustrated in the drawings and described in detail in the detailed description. However, this is not intended to limit the scope to the specific embodiments, and should be understood to include various modifications, equivalents, and/or alternatives of the embodiment of the disclosure. In connection with the description of the drawings, like reference numerals may be used for like elements.

[0044] In describing the disclosure, when it is determined that a detailed description of a related known function or configuration may unnecessarily obscure the gist of the disclosure, a detailed description thereof will be omitted.

[0045] In addition, the following embodiments may be modified in many different forms, and the scope of the technical idea of the disclosure is not limited to the following embodiments. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the spirit of the disclosure to those skilled in the art.

[0046] Terms used in this disclosure are only used to describe specific embodiments, and are not intended to limit the scope of rights. Singular expressions include plural expressions unless the context clearly dictates otherwise.

[0047] In this disclosure, expressions such as “has,” “can have,” “includes,” or “can include” indicate the existence of a corresponding feature (e.g., numerical value, function, operation, or component such as a part) and do not preclude the existence of additional features.

[0048] In this disclosure, expressions such as “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may include all possible combinations of the items listed together. For example, “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may refer to all cases (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

[0049] Expressions such as “first,” “second,” “primary,” or “secondary,” as used in this disclosure may modify various components regardless of order and/or importance, are used only to distinguish one component from other components, and do not limit the corresponding components.

[0050] Further, terms such as ‘leading end’, ‘rear end’, ‘upper side’, ‘lower side’, ‘top end’, ‘bottom end’, etc. used in the disclosure are defined with reference to the drawings. However, the shape and position of each component are not limited by these terms.

[0051] Hereinafter, embodiments of a refrigerator 1 according to the disclosure will be described in detail with reference to the attached drawings.

[0052] FIG. 1 is a rear perspective view illustrating a refrigerator 1 according to an embodiment of the disclosure.

[0053] Referring to FIG. 1, a refrigerator 1 according to an embodiment of the disclosure may include a main body 10.

[0054] The main body 10 may form the exterior of the refrigerator 1. A refrigerating compartment and a freezing compartment for accommodating food may be provided inside the main body 10.

[0055] A refrigerating compartment opening communicating with the refrigerating compartment and a freezing compartment opening communicating with the freezing com-

partment may be provided on the front side of the main body 10. In addition, a refrigerating compartment door and a freezing compartment door for opening and closing the refrigerating compartment opening and the freezing compartment opening may be provided on the front side of the main body 10.

[0056] The main body 10 may be provided with a refrigeration cycle device capable of cooling the refrigerating compartment and the freezing compartment.

[0057] The refrigeration cycle device may include a compressor, a condenser, an expansion valve, and a heat exchanger.

[0058] The heat exchanger may be disposed adjacent to the refrigerating compartment and the freezing compartment.

[0059] The compressor, the condenser, and the expansion valve may be disposed in a machine room 20 provided at a lower portion of the main body 10. In other words, the machine room 20 may be provided at the lower portion of the main body 10, and the compressor, the condenser, and the expansion valve may be disposed in the machine room 20.

[0060] The machine room 20 may be covered by a cover 40 that is detachable. Therefore, when the cover 40 is opened, the machine room 20 is exposed, thereby allowing the worker to access the machine room 20.

[0061] Hereinafter, the machine room 20 of the refrigerator 1 according to an embodiment of the disclosure will be described in detail with reference to FIGS. 2 to 5.

[0062] FIG. 2 is a partial enlarged perspective view illustrating the refrigerator 1 of FIG. 1. FIG. 3 is a perspective view illustrating a state in which a cover 40 is separated from the refrigerator 1 of FIG. 2. FIG. 4 is a partial perspective view illustrating a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure. FIG. 5 is an enlarged perspective view of part A of FIG. 4.

[0063] Referring to FIGS. 2 to 4, the machine room 20 may be provided at the lower portion of the main body 10 of the refrigerator 1. In other words, the machine room 20 may be provided adjacent to the bottom of the refrigerator 1, that is, the bottom of the main body 10. The machine room 20 may be formed of a chassis and have a space having a roughly rectangular parallelepiped shape.

[0064] The compressor 11, the condenser, and the expansion valve may be disposed in the machine room 20. In the machine room 20 of FIGS. 3 and 4, only the compressor 11 that generates vibration is shown for the convenience of illustration.

[0065] An opening 21 communicating with the machine room 20 may be provided on one side of the main body 10. In the case of this embodiment, the opening 21 may be provided on the rear side of the main body 10. The worker may access the compressor 11, the condenser, and the expansion valve disposed in the machine room 20 through the opening 21.

[0066] The opening 21 may be formed in a shape corresponding to the machine room 20. For example, the opening 21 of the machine room 20 may be formed in an approximately rectangular shape.

[0067] The opening 21 of the machine room 20 may be covered with the cover 40. In other words, the cover 40 may be disposed on one side of the main body 10 so as to cover the opening 21 of the machine room 20.

[0068] The machine room 20 may include a support part 22 supporting the cover 40. The support part 22 may be provided to contact the cover 40 at the edge of the opening 21 of the machine room 20. The support part 22 may be formed along the entire circumference of the opening 21. The outer surface of the support part 22 in contact with the cover 40 may be formed in a plane.

[0069] The support part 22 may be formed as a part of the chassis forming the machine room 20. For example, the part of the chassis forming the opening 21 of the machine room 20 may form the support part 22.

[0070] The support part 22 of the machine room 20 may include an upper support part 221, a lower support part 222, a left support part 223, and a right support part 224.

[0071] The upper support part 221 may be provided on the upper side of the opening 21. A plurality of screw holes 23 may be provided on the upper support part 221. In other words, the plurality of screw holes 23 may be provided at the upper side of the opening 21 on one side of the main body 10. A female screw may be provided on the inner surface of each of the plurality of screw holes 23.

[0072] The left support part 223 may be provided on the left side of the opening 21. The right support part 224 may be provided on the right side of the opening 21.

[0073] The lower support part 222 may be provided on the lower side of the opening 21. A plurality of main body hooks 30 may be provided on the lower support part 222. In other words, the plurality of main body hooks 30 may be provided below the opening 21 on one side of the main body 10. The plurality of main body hooks 30 may be formed to support the cover 40.

[0074] The plurality of main body hooks 30 may be disposed at a certain interval on the lower support part 222. Each of the plurality of main body hooks 30 may be formed to be inclined upward on one side of the main body 10. The plurality of main body hooks 30 may be formed below the lower end of the opening 21.

[0075] The plurality of main body hooks 30 may be formed in the same shape, so one main body hook 30 will be described below.

[0076] Referring to FIG. 5, the main body hook 30 may include a hook portion 31 and a guide portion 32.

[0077] The hook portion 31 may protrude outward from one side of the main body 10. The hook portion 31 may be formed approximately perpendicular to one side of the main body 10. The hook portion 31 may be formed approximately in a rectangular shape. The hook portion 31 may be formed with the same thickness as the chassis forming the main body 10.

[0078] The guide portion 32 may extend upward from the tip of the hook portion 31. The guide portion 32 may be formed so that the upper end of the guide portion 32 is positioned lower than the lower end of the opening 21.

[0079] The guide portion 32 may be formed so that the width thereof becomes narrower from the bottom to the top. In other words, the width of the lower end of the guide portion 32 connected to the hook portion 31 may be larger than the width of the upper end of the guide portion 32. For example, the guide portion 32 may be formed in an approximately triangular shape.

[0080] The guide portion 32 may be formed so as to be inclined upward from the tip of the hook portion 31. When

the guide portion 32 is formed so as to be inclined upward, the cover 40 may be easily coupled to the main body hook 30.

[0081] The main body hook 30 may be formed integrally with the main body 10. For example, the main body hook 30 may be formed by press-processing the lower support part 222 of the machine room 20 formed on the main body 10.

[0082] The cover 40 may be coupled to the plurality of main body hooks 30. The cover 40 may be formed to cover the opening 21 of the machine room 20 of the main body 10.

[0083] Hereinafter, the cover 40 will be described in detail with reference to FIGS. 6 to 8.

[0084] FIG. 6 is a perspective view illustrating a cover 40 configured to cover an opening 21 of a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure. FIG. 7 is a perspective view illustrating the cover 40 of FIG. 6 as viewed from the rear. FIG. 8 is an enlarged perspective view of part B of FIG. 7.

[0085] Referring to FIGS. 6 and 7, the cover 40 may be formed as a flat plate having a thin thickness and a roughly rectangular shape. The cover 40 may be formed of an elastic material. For example, the cover 40 may be formed of a thin metal plate.

[0086] The cover 40 may include a cover portion 41 and a fixing portion 42.

[0087] The cover portion 41 may be formed in a roughly rectangular shape so as to cover the opening 21 of the machine room 20. The cover portion 41 may include a plurality of air holes 45. The plurality of air holes 45 may be formed as slits. External air may be introduced into the machine room 20 through the plurality of air holes 45.

[0088] The fixing portion 42 may be provided at the edge of the cover portion 41. The fixing portion 42 may be formed along the entire circumference of the cover portion 41. The fixing portion 42 may be formed to correspond to the support part 22 of the machine room 20.

[0089] In order to increase the rigidity of the cover 40, the fixing portion 42 and the cover portion 41 may be formed to have a step. For example, the cover portion 41 may be formed as an approximately rectangular groove protruding from the fixing portion 42.

[0090] The fixing portion 42 may include an upper fixing portion 421, a lower fixing portion 422, a left fixing portion 423, and a right fixing portion 424.

[0091] The upper fixing portion 421 may be provided on the upper portion of the cover 40, that is, on the upper side of the cover portion 41. The upper fixing portion 421 may be formed to correspond to the upper support part 221 of the machine room 20. The upper fixing portion 421 may contact the upper support part 221 of the machine room 20.

[0092] The upper fixing portion 421 may include a plurality of bolt holes 43 corresponding to the plurality of screw holes 23 of the upper support part 221. Accordingly, the upper fixing portion 421 of the cover 40 may be fixed to the upper support part 221 of the main body 10 with a plurality of bolts.

[0093] The left fixing portion 423 may be provided on the left side of the cover portion 41. The right fixing portion 424 may be provided on the right side of the cover portion 41.

[0094] The lower fixing portion 422 may be provided on the lower portion of the cover 40, that is, on the lower side of the cover portion 41. The lower fixing portion 422 may be formed to correspond to the lower support part 222 of the

machine room 20. The lower fixing portion 422 may contact the lower support part 222 of the machine room 20.

[0095] The lower fixing portion 422 may include a plurality of cover catches 50. In other words, the plurality of cover catches 50 may be provided at the lower portion of the cover 40. The plurality of cover catches 50 may be provided below the cover portion 41.

[0096] The plurality of cover catches 50 may be provided to correspond to the plurality of main body hooks 30 provided on one side of the main body 10. In detail, the plurality of cover catches 50 may be formed so as to be caught on the plurality of main body hooks 30 provided on the lower support part 222 of the machine room 20.

[0097] The plurality of cover catches 50 may be formed in the same shape, so one cover catch 50 will be described below.

[0098] Referring to FIG. 8, the cover catch 50 may be formed in a protrusion shape protruding from one side of the cover 40. In other words, the cover catch 50 may be formed in a protrusion shape protruding from the lower fixing portion 422 of the cover 40 toward the machine room 20. Accordingly, the cover catch 50 may protrude from one surface of the fixing portion 42 in a direction opposite to the cover portion 41.

[0099] A catch groove 53 may be provided at the lower end 51 of the cover catch 50. The catch groove 53 of the cover catch 50 may be formed so that the main body hook 30 is inserted into the catch groove 53. In other words, the guide portion 32 of the main body hook 30 may be inserted into the catch groove 53 of the cover catch 50.

[0100] In the case of the embodiment shown in FIG. 8, the cover catch 50 may be formed in a U-shape with a substantially flat bottom. In other words, the cover catch 50 may be formed in a channel shape. The upper end 52 and the lower end 51 of the cover catch 50 may be open. The open lower end 51 of the cover catch 50 may form the catch groove 53. Therefore, when the cover 40 is disposed in the opening 21 of the machine room 20, the guide portion 32 of the main body hook 30 may be inserted into the catch groove 53 of the cover catch 50.

[0101] The inner width W2 of the cover catch 50 may be formed in a size corresponding to the width W1 of the main body hook 30 (see FIG. 5). In other words, the inner width W2 of the cover catch 50 may be formed to be larger than the width W1 of the main body hook 30.

[0102] The left-right movement amount of the cover 40 may be defined by the cover catches 50 and the main body hooks 30. In other words, the left-right movement amount of the cover 40 may be defined by the difference between the inner width W2 of the cover catch 50 and the width W1 of the main body hook 30.

[0103] FIG. 9 is an enlarged perspective view of part C of FIG. 2. FIG. 10 is a cross-sectional view taken along line C-C of FIG. 9.

[0104] Referring to FIGS. 9 and 10, the cover catch 50 may be coupled to the main body hook 30. In detail, the guide portion 32 of the main body hook 30 may be inserted into the catch groove 53 of the cover catch 50, and the lower end 51 of the cover catch 50 may be supported by the hook portion 31 of the main body hook 30. In addition, the cover catch 50 may be located between the guide portion 32 of the main body hook 30 and the lower support part 222 of the machine room 20.

[0105] Accordingly, the left-right movement of the cover 40 may be limited by the guide portion 32 of the main body hook 30, and the downward movement of the cover 40 may be limited by the hook portion 31 of the main body hook 30. In addition, the forward-backward movement of the cover 40, that is, the movement of the cover 40 in the direction perpendicular to the rear side of the refrigerator 1, may be limited by the guide portion 32 of the main body hook 30.

[0106] When the cover 40 is assembled to the machine room 20, that is, when the cover catches 50 of the cover 40 are coupled to the main body hooks 30 of the machine room 20, there may be a gap G between the lower fixing portion 422 of the cover 40 and the lower support part 222 of the machine room 20. The gap G between the cover 40 and the support part 22 of the machine room 20 may be defined by the cover catches 50 and the main body hooks 30. For example, the gap G between the cover 40 and the support part 22 of the machine room 20 may be defined to be the same as the height H1 of the cover catch 50.

[0107] The cover 40 may include an emboss 60. The emboss 60 may be provided on the lower portion of the cover 40, that is, the lower fixing portion 422. The emboss 60 may be provided below the cover portion 41.

[0108] The emboss 60 may be formed to protrude from one surface of the cover 40 in the same direction as the plurality of cover catches 50.

[0109] The emboss 60 may be provided between the plurality of cover catches 50. The emboss 60 may be provided between two adjacent cover catches 50 among the plurality of cover catches 50.

[0110] As another example, the emboss 60 may include a plurality of embosses 60. For example, when the cover 40 includes four cover catches 50, three embosses 60 may be formed between the four cover catches 50. When the plurality of embosses 60 are formed, the plurality of embosses 60 may be formed to form a straight line.

[0111] The emboss 60 may include a lower inclined surface 61 that faces the lower end of the cover 40 and is formed to be inclined upward. In detail, one surface of the emboss 60 that faces the lower end of the cover 40 may be formed to be inclined upward with respect to the cover 40. In other words, the emboss 60 may include the lower inclined surface 61 that is inclined upward with respect to the cover 40.

[0112] For this purpose, the emboss 60 may be formed to have an arc cross-section or a triangular cross-section. In other words, the emboss 60 may be formed as a column having an arc cross-section or a triangular column. However, the shape of the emboss 60 is not limited thereto. The emboss 60 may be formed as a column having various cross-sections having the lower inclined surface 61.

[0113] As described above, in the case where the emboss 60 includes the lower inclined surface 61, when assembling the cover 40 to the machine room 20, the emboss 60 may be prevented from interfering with the lower support part 222 of the machine room 20.

[0114] The emboss 60 may be formed higher than the height H1 of the plurality of cover catches 50. In other words, the height H2 of the emboss 60 may be higher than the height H1 of the cover catches 50. When the height H2 of the emboss 60 is made higher than the height H1 of the cover catches 50, the tip of the emboss 60 may be in close contact with the support part 22 of the machine room 20. Accordingly, it is possible to prevent a gap from being

formed between the support part 22 of the machine room 20 and the fixing portion 42 of the cover 40.

[0115] FIG. 11 is a cross-sectional view taken along line D-D of FIG. 9.

[0116] Referring to FIG. 11, when the cover catch 50 of the cover 40 is caught on the main body hook 30 of the machine room 20, the tip of the emboss 60 may be in contact with the lower support part 222 of the machine room 20. At this time, because the height H2 of the emboss 60 is greater than the gap G between the cover 40 and the support part 22 of the machine room 20, that is, the lower fixing portion 422 of the cover 40 and the lower support part 222 of the machine room 20, the emboss 60 may be in close contact with the support part 22 of the machine room 20.

[0117] In other words, the cover 40 may be pressed against the support part 22 of the machine room 20 with a force applied to it by the difference between the gap G between the cover 40 and the support part 22 of the machine room 20 and the height H2 of the emboss 60. Therefore, there is no gap between the cover 40 and the support part 22 of the machine room 20, that is, between the tip of the emboss 60 of the cover 40 and the lower support part 222 of the machine room 20.

[0118] The cover catches 50 and the emboss 60 may be formed integrally with the cover 40. For example, when manufacturing the cover 40, the lower fixing portion 422 may be press-processed to form the plurality of cover catches 50 and the emboss 60.

[0119] The lower portion of the cover 40 may be fixed to the lower support part 222 of the machine room 20 by the cover catches 50 described above, and the upper portion of the cover 40 may be fixed to the upper support part 221 of the machine room 20 by the fastening members.

[0120] In the case of this embodiment, a plurality of bolts may be used as the fastening members. In other words, by inserting the plurality of bolts into the plurality of bolt holes 43 of the cover 40 and fastening them to the plurality of screw holes 23 provided in the support part 22 of the machine room 20, the upper portion of the cover 40 may be fixed to the main body 10 above the opening 21 of the machine room 20.

[0121] However, the fastening member that secures the upper portion of the cover 40 to the upper support part 221 of the machine room 20 is not limited thereto. Any type of fastening member may be used as long as the upper portion of the cover 40 can be secured to the upper support part 221 of the machine room 20. For example, a one-touch coupling structure may be used as the fastening member.

[0122] Hereinafter, a method of assembling the cover 40 to the machine room 20 of the main body 10 according to an embodiment of the disclosure will be described with reference to FIGS. 12A to 12C.

[0123] FIG. 12A is a partial cross-sectional view illustrating a state before assembling a cover 40 to a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure. FIG. 12B is a partial cross-sectional view illustrating a state in which a cover catch 50 of a cover 40 is coupled to a main body hook 30 of a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure. FIG. 12C is a partial cross-sectional view illustrating a state in which a cover 40 is assembled to a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure.

[0124] Referring to FIG. 12A, the cover 40 may be positioned obliquely in front of the machine room 20 of the main body 10. At this time, the cover 40 may be positioned so that the plurality of cover catches 50 face downward. In addition, the cover 40 may be positioned so that the plurality of cover catches 50 of the cover 40 correspond to the plurality of main body hooks 30 of the main body 10.

[0125] In the state of FIG. 12A, the cover 40 may be moved downward in the direction of the arrow. In other words, the main body hooks 30 of the main body 10 may be obliquely inserted into the cover catches 50 of the cover 40. Then, as illustrated in FIG. 12B, the plurality of cover catches 50 of the cover 40 may be caught on the plurality of main body hooks 30 of the main body 10. In detail, the lower ends of the plurality of cover catches 50 of the cover 40 may be positioned on the hook portions 31 of the main body hooks 30 of the main body 10. In other words, the plurality of cover catches 50 may be positioned between the lower support part 222 of the main body 10 and the guide portions 32 of the main body hooks 30.

[0126] In other words, the upper and lower positions and the left and right positions of the cover 40 may be defined by the plurality of main body hooks 30 of the machine room 20 and the plurality of cover catches 50 of the cover 40.

[0127] In the state of FIG. 12B, the upper portion of the cover 40 may be moved to the left in the direction of the arrow. In other words, the upper portion of the cover 40 may be pushed toward the machine room 20. Then, as illustrated in FIG. 12C, the upper fixing portion 421 of the cover 40 may come into contact with the upper support part 221 of the machine room 20.

[0128] In this state, when the cover 40 is fixed to the main body 10 with the plurality of fastening members, the assembly of the cover to the front side of the machine room 20 of the main body 10 may be completed. For example, the upper fixing portion 421 of the cover 40 may be fixed to the upper support part 221 of the main body 10 using the plurality of bolts.

[0129] As described above, when the cover 40 is assembled to the support part 22 of the main body 10, the cover 40 is in close contact with the support part 22 of the main body 10, so that there may be no gap between the cover 40 and the support part 22 of the main body 10. Hereinafter, the force acting between the lower fixing portion 422 of the cover 40 and the lower support part 222 of the main body 10 in the state where the cover 40 is assembled to the main body 10 will be described in detail with reference to FIG. 13.

[0130] FIG. 13 is a view illustrating forces acting between a machine room 20 and a cover 40 when the cover 40 is disposed in the machine room 20 of a refrigerator 1 according to an embodiment of the disclosure.

[0131] Referring to FIG. 13, the lower end 51 of the cover catch 50 may be caught on the main body hook 30, that is, a portion connecting the hook portion 31 and the guide portion 32, so that the cover catch 50, that is, the cover 40, may be applied with force in the direction of arrow F1.

[0132] In addition, because the tip of the emboss 60 of the cover 40 is in contact with the lower support part 222 of the main body 10, force may be applied to the emboss 60, that is, the cover 40, in the direction of arrow F2. In detail, because the height H2 of the emboss 60 is greater than the gap G between the lower fixing portion 422 of the cover 40 and the lower support part 222 of the main body 10 or the

height H1 of the cover catch 50, force may be applied to the emboss 60 of the cover 40 in the direction of arrow F2.

[0133] Accordingly, the cover 40 may be forcibly fitted to the main body hooks 30 of the main body 10 by the force in the F1 direction applied to the cover catch 50 and the force in the F2 direction applied to the emboss 60. Therefore, no gap may occur between the emboss 60 of the cover 40 and the lower support part 222 of the main body 10.

[0134] The compressor 11 (see FIG. 4) may be disposed in the machine room 20. Accordingly, when the compressor 11 operates, vibration may occur. This vibration may be transmitted to the lower support part 222 through the chassis forming the machine room 20 in which the compressor 11 is disposed.

[0135] The opening 21 of the machine room 20 may be covered with the cover 40. Therefore, the lower support part 222 of the machine room 20 may face the lower fixing portion 422 of the cover 40. At this time, when there is a gap between the lower support part 222 of the main body 10 and the cover 40, noise may be generated as the cover 40 vibrates slightly due to the vibration of the compressor 11.

[0136] However, in the refrigerator 1 according to an embodiment of the disclosure, because the lower support part 222 of the machine room 20 and the lower fixing portion 422 of the cover 40 are forcibly fitted by the emboss 60 and the cover catches 50, noise in which the cover 40 vibrates slightly due to the vibration of the compressor 11, that is, vibration noise may not occur.

[0137] In general, in order to prevent the vibration noise of the cover 40, a vibration-proof member may be disposed between the cover 40 and the main body 10. However, because the vibration-proof member is formed of a material such as sponge, rubber, and the like, there may be a problem that the gap between the cover 40 and the main body 10 may not be maintained uniformly. When the gap between the cover 40 and the main body 10 is not uniform, the vibration noise of the cover 40 may occur.

[0138] In addition, when the vibration-proof member is used, there may be a problem that the material cost of the vibration-proof member and the cost of attaching the vibration-proof member to the cover 40 are incurred.

[0139] Alternatively, the cover 40 may be coupled to the main body 10 using a plurality of bolts to prevent the vibration noise of the cover 40. However, when the threads of the bolts are broken during assembly, a gap may occur between the cover 40 and the main body 10. In addition, when the positions of the screw holes of the main body 10 and the positions of the bolt holes of the cover 40 do not match, a gap may occur between the cover 40 and the main body 10. When the gap occurs between the cover 40 and the main body 10, the vibration noise of the cover 40 may occur.

[0140] In addition, when a number of bolts are used, there may be a problem that the material cost and assembly cost are incurred as much as the number of bolts.

[0141] However, in the case of the disclosure, the cover 40 may be tightly attached to the main body 10 without a gap by using the plurality of main body hooks 30 provided on the main body 10 and the plurality of cover catches 50 and emboss 60 provided on the cover 40. Therefore, the refrigerator 1 according to an embodiment of the disclosure may reduce the material cost and assembly cost of the cover 40 compared to the refrigerator according to the prior art.

[0142] In the above, the case where the emboss 60 is provided on the lower fixing portion 422 of the cover 40 has

been described. However, the disclosure is not limited thereto. Conversely, the emboss 60 may be provided on the lower support part 222 of the main body 10. Hereinafter, a refrigerator 1 in which the emboss 60 is provided on the main body 10 will be described with reference to FIGS. 14 and 15.

[0143] FIG. 14 is a perspective view illustrating another example of a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure. FIG. 15 is a perspective view illustrating a cover 40 disposed in the machine room 20 of the refrigerator 1 of FIG. 14.

[0144] Referring to FIG. 14, the lower support part 222 may be provided at the lower side of the opening 21 of the machine room 20. The plurality of main body hooks 30 may be provided on the lower support part 222. In other words, the plurality of main body hooks 30 may be provided below the opening 21 on one side of the main body 10. The plurality of main body hooks 30 may be formed to support the cover 40.

[0145] The plurality of main body hooks 30 may be disposed at a certain interval on the lower support part 222. The plurality of main body hooks 30 may be formed in the same shape.

[0146] The main body hook 30 may be formed in a protrusion shape protruding from one side of the main body 10. In other words, the main body hook 30 may be formed in a protrusion shape protruding outward from the lower support part 222 of the main body 10.

[0147] A catch groove 33 may be provided at the upper end of the main body hook 30. The catch groove 33 of the main body hook 30 may be formed so that a cover catch 50 is inserted into the catch groove 33.

[0148] In the case of the embodiment shown in FIG. 14, the main body hook 30 may be formed in a U-shape with a substantially flat bottom. In other words, the main body hook 30 may be formed in a channel shape. The upper end 34 and lower end 35 of the main body hook 30 may be open. The open upper end 34 of the main body hook 30 may form the catch groove 33. Therefore, when disposing the cover 40 in the opening 21 of the machine room 20, the cover catch 50 may be inserted into the catch groove 33 of the main body hook 30.

[0149] An emboss 60 may be provided on the lower support part 222 of the main body 10. In other words, the emboss 60 may be provided below the opening 21 of the machine room 20.

[0150] The emboss 60 may be formed to protrude from one side of the main body 10 in the same direction as the plurality of main body hooks 30.

[0151] The emboss 60 may be provided between the plurality of main body hooks 30. The emboss 60 may be provided between two adjacent main body hooks 30 among the plurality of main body hooks 30.

[0152] As another example, the emboss 60 may include a plurality of embosses 60. For example, when four main body hooks 30 are provided on the main body 10, three embosses 60 may be formed between the four main body hooks 30. When the plurality of embosses 60 are provided, the plurality of embosses 60 may be formed to form a straight line.

[0153] The emboss 60 may include an upper inclined surface 62 that faces the lower end of the opening 21 and is formed to be inclined downward. In detail, one surface of the emboss 60 that faces the lower end of the opening 21 may be formed to be inclined downward with respect to the lower

support part 222 of the main body 10. In other words, the emboss 60 may include the upper inclined surface 62 that is inclined downward with respect to the lower support part 222 of the main body 10.

[0154] For this purpose, the emboss 60 may be formed to have an arc cross-section or a triangular cross-section. In other words, the emboss 60 may be formed as a column having an arc cross-section or a triangular column. However, the shape of the emboss 60 is not limited thereto. The emboss 60 may be formed as a column having various cross-sections having the upper inclined surface 62.

[0155] As described above, in the case where the emboss 60 includes the upper inclined surface 62, when assembling the cover 40 to the machine room 20, the emboss 60 may be prevented from interfering with the lower end of the cover 40.

[0156] The emboss 60 may be formed higher than the height of the plurality of main body hooks 30. In other words, the height of the emboss 60 may be higher than the height of the main body hooks 30. When the height of the emboss 60 is made higher than the height of the main body hooks 30, the emboss 60 may be in close contact with the fixing portion 42 of the cover 40. Accordingly, it is possible to prevent a gap from being formed between the support part 22 of the machine room 20 and the fixing portion 42 of the cover 40.

[0157] Referring to FIG. 15, the cover 40 may include a cover portion 41 and a fixing portion 42.

[0158] The cover portion 41 may be formed in a roughly rectangular shape so as to cover the opening 21 of the machine room 20. The cover portion 41 may include a plurality of air holes 45. The plurality of air holes 45 may be formed as slits. External air may be introduced into the machine room 20 through the plurality of air holes 45.

[0159] The fixing portion 42 may be provided at the edge of the cover portion 41. The fixing portion 42 may be formed along the entire circumference of the cover portion 41. The fixing portion 42 may be formed to correspond to the support part 22 of the machine room 20.

[0160] In order to increase the rigidity of the cover 40, the fixing portion 42 and the cover portion 41 may be formed to have a step. For example, the cover portion 41 may be formed as an approximately rectangular groove protruding from the fixing portion 42.

[0161] The fixing portion 42 may include an upper fixing portion 421, a lower fixing portion 422, a left fixing portion 423, and a right fixing portion 424.

[0162] The upper fixing portion 421, the left fixing portion 423, and the right fixing portion 424 are the same as the above-described embodiment, so detailed descriptions thereof are omitted.

[0163] The lower fixing portion 422 may be provided on the lower portion of the cover 40, that is, on the lower side of the cover portion 41. The lower fixing portion 422 may be formed to correspond to the lower support part 222 of the machine room 20. The lower fixing portion 422 may be in contact with the lower support part 222 of the machine room 20.

[0164] The lower fixing portion 422 may include a plurality of a cover catches 50. In other words, the plurality of cover catches 50 may be provided at the lower portion of the cover 40. The plurality of cover catches 50 may be provided below the cover portion 41.

[0165] The plurality of cover catches **50** may be provided to correspond to the plurality of main body hooks **30** provided on one side of the main body **10**. In detail, the plurality of cover catches **50** may be formed so as to be caught on the plurality of main body hooks **30** provided on the lower support part **222** of the machine room **20**.

[0166] The plurality of cover catches **50** may be provided at regular intervals on the lower fixing portion **422** of the cover **40**. The plurality of cover catches **50** may be formed in the same shape.

[0167] Referring to FIG. **15**, the cover catch **50** may include a catch portion **54** and a guide portion **55**.

[0168] The catch portion **54** may protrude from one side of the cover **40** toward the machine room **20** of the main body **10**. The catch portion **54** may be formed approximately perpendicular to one side of the cover **40**. The catch portion **54** may be formed in an approximately rectangular shape. The catch portion **54** may be formed with the same thickness as the cover **40**.

[0169] The guide portion **55** may extend downward from the tip of catch portion **54**. The guide portion **55** may be formed so that the lower end of the guide portion **55** is positioned above the lower end of the cover **40**.

[0170] The guide portion **55** may be formed so that the width of the guide portion **55** becomes narrower from the upper end to the lower end. In other words, the width of the upper end of the guide portion **55** connected to the catch portion **54** may be larger than the width of the lower end of the guide portion **55**. For example, the guide portion **55** may be formed in an approximately triangular shape.

[0171] The guide portion **55** may be formed to be inclined downward from the tip of the catch portion **54**. When the guide portion **55** is formed to be inclined downward, the cover catch **50** may be easily connected with the main body hook **30**.

[0172] When the guide portion **55** of the cover catch **50** of the cover **40** is inserted into the catch groove **33** of the main body hook **30** of the main body **10**, the cover catch **50** and the main body hook **30** may be coupled. In this state, when the upper fixing portion **421** of the cover **40** is fixed to the upper support part **221** of the main body **10** using a fastening member, the cover **40** may be assembled to the machine room **20** of the main body **10**.

[0173] Because there is the emboss **60** between the lower fixing portion **422** of the cover **40** and the lower support part **222** of the main body **10**, the lower fixing portion **422** of the cover **40** may be in close contact with the lower support part **222**. Accordingly, there may be no gap between the lower fixing portion **422** of the cover **40** and the lower support part **222** of the main body **10**. Therefore, when the compressor **11** operates, the vibration noise of the cover **40** may not occur.

[0174] As described above, the refrigerator **1** according to the disclosure may have the plurality of main body hooks **30** provided on one side of the main body **10** in which the machine room **20** is provided, the plurality of cover catches **50** provided on one side of the cover **40**, and the emboss **60** provided between the cover **40** and the main body **10**.

[0175] When the main body hooks **30** of the main body **10** and the cover catches **50** of the cover **40** are coupled, force may be applied to the cover **40** by the emboss **60** provided between the main body **10** and the cover **40**. Then, the cover **40** may be in close contact with the main body **10** so that there is no gap between the main body **10** and the cover **40**. In detail, when the emboss **60** is provided on the cover **40**,

the tip of the emboss **60** is in close contact with the main body **10**, so that the vibration noise of the cover **40** may be prevented. When the emboss **60** is provided on the main body **10**, the tip of the emboss **60** is in close contact with the cover **40**, so that the vibration noise of the cover **40** may be prevented.

[0176] In the refrigerator **1** according to the disclosure having the structure as described above, the cover **40** is disposed in close contact with one side of the main body **10** in which the machine room **20** is disposed, so that the vibration noise of the cover **40** caused by the vibration of the compressor **11** may be prevented or minimized.

[0177] In addition, because the refrigerator **1** according to the disclosure may prevent the vibration noise by the cover **40** itself, the cost of the vibration-proof of the cover **40** may be reduced.

[0178] In the foregoing, the disclosure has been shown and described with reference to various embodiments. However, it is understood by those skilled in the art that various changes may be made in form and detail without departing from the scope of the disclosure as defined by the appended claims and equivalents thereof.

1. A refrigerator comprising:

- a machine room to be provided at a lower portion of a main body;
- an opening to be provided on one side of the main body to allow access to the machine room;
- a cover to be disposed on the one side of the main body to cover the opening;
- a plurality of cover catches to be provided on a lower portion of the cover;
- an emboss to be provided between the plurality of cover catches on the lower portion of the cover; and
- a plurality of main body hooks to be provided on the one side of the main body below the opening such that while the plurality of main body hooks are provided on the one side of the main body, locations of the plurality of main body hooks and the plurality of cover catches of the cover correspond to each other.

2. The refrigerator of claim 1, wherein

while the opening of the machine room is covered with the cover, the plurality of cover catches are caught on the plurality of main body hooks, and the emboss contacts the one side of the main body on which the plurality of main body hooks are formed.

3. The refrigerator of claim 2, wherein

a height of the emboss is greater than a gap between the cover and the one side of the main body.

4. The refrigerator of claim 1, wherein

each of the plurality of main body hooks is formed to be inclined upwardly on the one side of the main body.

5. The refrigerator of claim 1, wherein

the emboss includes a lower inclined surface formed to be inclined upwardly and facing a lower end of the cover.

6. The refrigerator of claim 5, wherein

the emboss is formed to have an arc cross-section or a triangular cross-section.

7. The refrigerator of claim 1, wherein

the emboss is among a plurality of embosses arranged in a straight line.

8. The refrigerator of claim 1, wherein the cover is formed as a rectangular flat plate, each of the plurality of cover catches is formed in a protrusion shape protruding from one side of the cover, and
a catch groove into which a respective main body hook is insertable is provided at a lower end of a respective cover catch.
9. The refrigerator of claim 8, wherein the emboss is formed to protrude from one side of the cover along a direction that is same as a direction in which the plurality of cover catches protrude.
10. The refrigerator of claim 9, wherein a height of the emboss is higher than a height of at least one cover catch among the plurality of cover catches.
11. The refrigerator of claim 1, further comprising:
a fastening member formed to fasten an upper portion of the cover to the one side of the main body above the opening.
12. The refrigerator of claim 1, wherein the main body comprises an upper support part, a lower support part, a left support part, and a right support part provided at an edge of the opening,
the plurality of main body hooks are provided on the lower support part, and
a plurality of screw holes are provided on the upper support part.
13. The refrigerator of claim 12, wherein the cover comprises an upper fixing portion and a lower fixing portion,

- the plurality of cover catches are provided on the lower fixing portion, and
a plurality of bolt holes corresponding to the plurality of screw holes of the upper support part of the main body are provided on the upper fixing portion.
14. A refrigerator comprising:
a machine room to be provided at a lower portion of a main body;
an opening to be provided on one side of the main body to allow access to the machine room;
a cover to be disposed on the one side of the main body to cover the opening;
a plurality of main body hooks to be provided on the one side of the main body below the opening;
an emboss to be provided on one side of the main body between the plurality of main body hooks and below the opening; and
a plurality of cover catches to be provided on a lower portion of the cover such that while the cover is disposed on the one side of the main body, locations of the plurality of cover catches and the plurality of main body hooks of the main body correspond to each other.
15. The refrigerator of claim 14, wherein each of the plurality of cover catches is formed to be inclined downward on a lower portion of the cover,
each of the plurality of main body hooks is formed in a protrusion shape protruding from one side of the main body, and
a catch groove into which the cover catch is inserted is provided on an upper surface of the main body hook.

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